

Heterogeneous Decentralized Fine-Tuning of Large Language Models

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Abstract—Federated Learning (FL) enables privacy-preserving training of Large Language Models (LLMs) but struggles with system heterogeneity, particularly when clients possess varying computational capabilities. Standard aggregation methods require uniform model architectures, forcing a “lowest common denominator” approach that throttles high-end devices. We propose Subspace Projection Aggregation (SPA), a novel framework that addresses this by treating client updates as overlapping subspaces. Using Singular Value Decomposition (SVD), SPA projects high-rank updates from capable clients into optimal lower-rank approximations for resource-constrained peers. Experiments on the Yelp Review dataset using Qwen2.5-7B show that SPA achieves 63.8% accuracy, outperforming homogeneous baselines by 2.6% while reducing communication costs by 25%. Our approach effectively distills knowledge from heterogeneous sources, proving that device diversity can be an asset rather than a liability in federated LLM fine-tuning.

Index Terms—Federated Learning, Large Language Models, LoRA, Heterogeneity, SVD, Subspace Learning

I. INTRODUCTION

Large Language Models (LLMs) have fundamentally transformed Natural Language Processing, demonstrating emergent abilities across diverse tasks [1]–[3]. However, deploying them in sensitive domains like healthcare or finance creates significant privacy issues. Regulations like GDPR [4] and HIPAA prevent centralizing sensitive data, while recent studies have shown that LLMs can memorize training data, rendering them vulnerable to extraction attacks [5], [6].

Recent incidents underscore the critical importance of privacy-preserving approaches. In 2025, the Grok Chat Leak exposed over 370,000 private conversations through centralized server vulnerabilities, demonstrating the inherent risks of collecting and storing sensitive user data [7]. Such breaches highlight why data governance frameworks mandate strict controls on personal information processing.

Federated Learning (FL) offers a solution by keeping data on-device [8], [9]. However, standard FL faces a critical hurdle: **system heterogeneity** [10], [11]. In real-world networks, not all devices are equal; a high-end server may exist alongside a resource-constrained smartphone. When using Parameter-Efficient Fine-Tuning (PEFT) techniques like Low-Rank Adaptation (LoRA) [12], this leads to a LoRA rank mismatch. Traditional FL requires identical ranks, forcing the network to either exclude weaker devices (dropping data) or throttle powerful ones (wasting capacity) [13], [14].

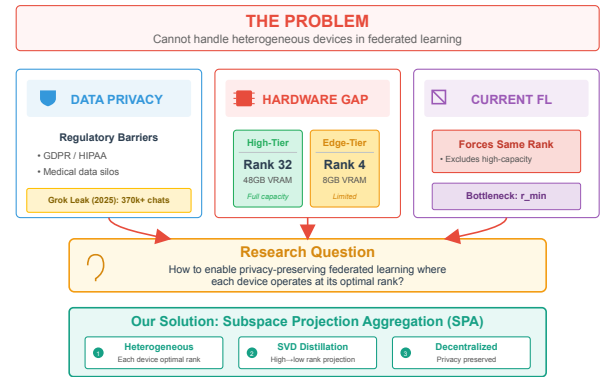


Fig. 1: **Heterogeneous Decentralized Fine-Tuning.** (1) *Data Governance* challenges [7]; (2) *Hardware Mismatch* between high-tier and edge-tier devices; and (3) *SPA Solution*: Reconstruct, SVD Analysis, and Project & Distill.

Figure 1 illustrates these challenges. The problem has three key dimensions: (1) *Data Governance* concerns; (2) *Hardware Mismatch*, where high-tier devices with 48GB VRAM can support LoRA rank 32 while edge-tier devices with only 8GB VRAM are limited to rank 4; and (3) the need for a robust aggregation strategy. While Zero-Padding allows mismatched ranks to be averaged, it introduces significant **truncation loss and sparsity noise**, diluting the global model’s intelligence.

To bridge this gap, we propose the SPA architecture. By utilizing Singular Value Decomposition (SVD), specifically leveraging the properties of the Eckart-Young-Mirsky theorem for optimal low-rank approximation [15], SPA treats updates as subspaces. This ensures that knowledge from high-rank clients is distilled into low-rank clients without the artifacts of padding. As shown in Figure 1, our approach operates through three key phases: (1) **Reconstruct** the full weight updates $\Delta W = B \times A$, (2) perform **SVD Analysis** to decompose the aggregated update as $W = U \Sigma V^T$, and (3) **Project & Distill** via dynamic rank mapping to extract principal subspaces tailored to each client’s hardware constraints.

II. BACKGROUND AND RELATED WORK

A. Federated Learning for Large Language Models

Federated Learning was originally proposed by McMahan et al. [8] for training neural networks on mobile devices. While effective for smaller models, scaling this approach to LLMs presents distinct challenges regarding communication overhead and memory constraints [16], [17].

Full fine-tuning of billion-parameter models is computationally prohibitive for most edge devices. Additionally, LLMs are susceptible to catastrophic forgetting. Recent efforts to adapt FL for LLMs, such as FedIT [18] and FATE-LLM [19], typically utilize adapter-based training. However, these approaches usually assume homogeneous client architectures, which limits their applicability in realistic, heterogeneous environments [20], [21].

B. Low-Rank Adaptation (LoRA) and PEFT

To mitigate the cost of fine-tuning, Parameter-Efficient Fine-Tuning (PEFT) methods have gained traction, including Adapters [22], Prefix Tuning [23], and P-Tuning [24]. Among these, LoRA [12] has become the de facto standard for consumer-grade hardware. It decomposes the weight update ΔW into two low-rank matrices, B and A . The forward pass is defined as:

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} B A x \quad (1)$$

where W_0 represents the frozen pre-trained weights.

Recent variants like QLoRA [25] integrate quantization to further reduce memory usage, while AdaLoRA [26] and DyLoRA [27] explore dynamic rank allocation during central training. However, applying these dynamic rank concepts in a *decentralized, federated* setting remains an open research problem.

C. Handling Heterogeneity in Federated Learning

System heterogeneity is a well-known challenge in FL. Existing solutions often employ **Client Selection** strategies, such as FedProx [11] and FedNova [28], which adjust aggregation based on client resources or training progress.

For model heterogeneity, HeteroFL [13] and Fjord [14] introduced the concept of training sub-models on edge devices. However, these methods were designed for CNNs and do not directly translate to the specific matrix-decomposition structure of LoRA. More recently, approaches like FlexLoRA [29], FLoRA [30], and SLoRA [31] have attempted to bridge this gap, though they often rely on complex stacking mechanisms or heuristic padding.

D. Singular Value Decomposition in Machine Learning

SVD factorizes a matrix W into three components ($W = U \Sigma V^T$). In Deep Learning, SVD has been extensively used for model compression [32], [33]. In our proposed framework, SVD serves as a spectral filter to remove noise from the aggregated model, similar to techniques used in subspace learning [34].

III. METHODOLOGY

A. Problem Formulation

We consider a federated network with K clients. Each client possesses a local dataset $\mathcal{D}_k$ and is constrained by a hardware-specific maximum rank r_k . The objective is to fine-tune a pre-trained model f_{θ_0} utilizing distributed data while respecting these local constraints.

Using the LoRA framework, the pre-trained weights W_0 remain frozen. Each client trains a pair of adapter matrices (A_k, B_k) . The intended weight update is:

$$\Delta W_k = B_k A_k \in \mathbb{R}^{d \times k} \quad (2)$$

The primary challenge is aggregation. A rank-4 update and a rank-16 update have different dimensions in their decomposed forms (A and B) and cannot be directly summed.

B. Subspace Projection Aggregation Algorithm

We address this with the SPA algorithm (Algorithm ??), which executes over T communication rounds.

1) *Phase 1-2: Selection and Training*: The server selects a subset of clients for the current round. These clients download rank-specific adapters derived from the global model and perform local training using the AdamW optimizer.

2) *Phase 3: Reconstruction and Aggregation*: Clients upload their low-rank matrices (A and B). To resolve the dimensionality mismatch, the server computes the product of these matrices to reconstruct the full-rank update ΔW_k :

$$\Delta W_k = B_k A_k \in \mathbb{R}^{d \times k} \quad (3)$$

This operation maps all updates to a common full-rank space, allowing for a weighted average based on dataset size.

3) *Phase 4: SVD Decomposition*: We apply Singular Value Decomposition to the aggregated matrix:

$$W_{agg} = U \Sigma V^T \quad (4)$$

Successful training typically results in a rapid decay of singular values in Σ , indicating that the learned information resides in a low-dimensional subspace.

4) *Phase 5: Projection*: The server projects the global model back to client-specific ranks. For a client with rank capacity r_j , we retain only the top- r_j components:

$$\begin{aligned} A_j &= \sqrt{\Sigma_{1:r_j}} \cdot V_{1:r_j}^T \\ B_j &= U_{:,1:r_j} \cdot \sqrt{\Sigma_{1:r_j}} \end{aligned} \quad (5)$$

Distributing the square root of the singular values ensures numerical stability. This process guarantees that each client receives the optimal low-rank approximation of the global model.

TABLE I: Comparison of Recent Works on Heterogeneous LoRA in Federated Fine-Tuning.

Work	Year	Hetero.	Aggregation Method	Limitation
FedAvg [8]	2017	No	Direct averaging	Assumes homogeneous clients
Zero-Padding	–	Yes	Zero-pad low-rank to max rank	Sparsity artifacts and noise
FLoRA [30]	2024	Yes	Stacking-based aggregation	Focuses on stacking
FlexLoRA [29]	2024	Yes	Full ΔW + SVD redist.	Similar motivation
HeLoRA [35]	2025	Yes	Context-based padding	Relies on padding variant
ILoRA [36]	2025	Yes	QR decomposition	Addresses initialization
SPA (Ours)	–	Yes	Full $\Delta W \rightarrow \text{SVD} \rightarrow \text{top-}r$	Optimal low-rank approx.

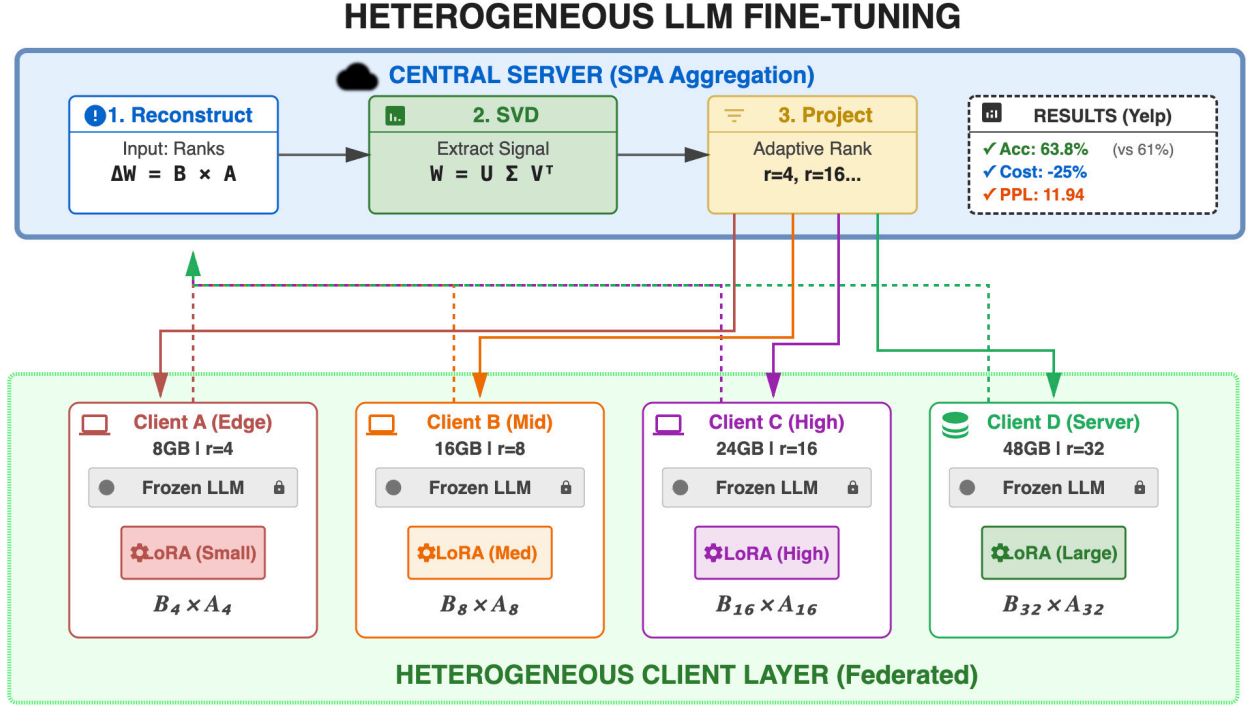


Fig. 2: System Architecture of Heterogeneous Decentralized Fine-Tuning with Subspace Projection Aggregation (SPA).

C. Theoretical Justification

Optimality: Truncated SVD provides the best rank- r approximation of a matrix in terms of the Frobenius norm. Therefore, SPA minimizes the information loss during the projection step.

Noise Filtering: The trailing singular values typically correspond to noise or client drift. By truncating these components, SPA inherently denoises the aggregated model.

D. Computational Complexity

The SVD operation on the server introduces computational overhead. However, for a layer dimension $d = 4096$, the decomposition takes approximately 2 to 5 seconds on a

modern GPU. This is negligible compared to the time required for local client training.

E. Baseline Comparisons and Rationale

We compare SPA against two primary categories of baselines:

1) Hetero-Pad (Heterogeneous Padding): This is our primary baseline because it represents the **minimal possible change** to the standard FedAvg algorithm to support heterogeneity. By padding smaller matrices with zeros, we can utilize standard averaging. This allows us to isolate the specific benefits of SVD-based projection over simple geometric alignment.

2) Homogeneous FedAvg ($r = 4$ and $r = 8$): We focus on $r = 4$ and $r = 8$ as the "lowest common denominator"

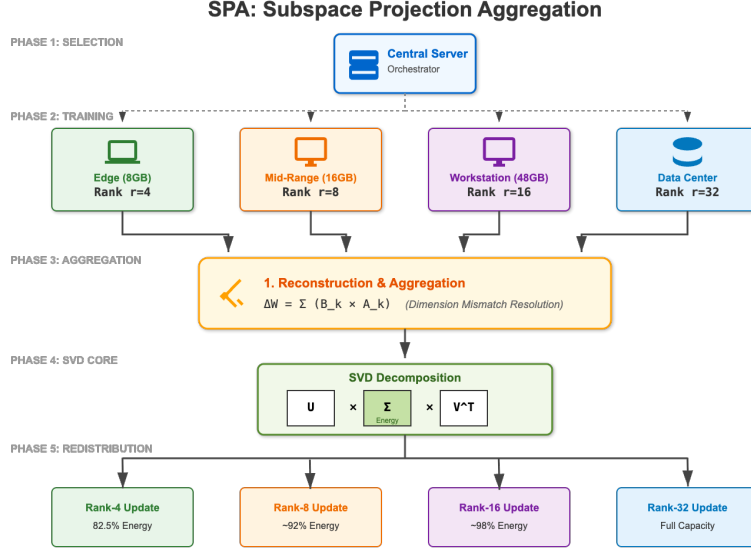


Fig. 3: Detailed workflow of Subspace Projection Aggregation across five phases.

1: Algorithm 1: Subspace Projection Aggregation (SPA)

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2: Require: Client ranks  $\{r_k\}_{k=1}^K$ , datasets  $\{\mathcal{D}_k\}_{k=1}^K$ 
3: Require: Pre-trained model weights  $W_0$ , number of rounds  $T$ 
4: Server Initialization:  $W_{global} \leftarrow 0$ 
5: for round  $t = 1, \dots, T$  do
6:   Phase 1: Client Selection
7:   Select a subset  $S_t$  of clients
8:   Phase 2: Download & Local Training
9:   for each client  $k \in S_t$  do
10:    Download  $(A_k^{(t)}, B_k^{(t)})$  (projected to rank  $r_k$ )
11:    Perform local training on  $\mathcal{D}_k$  for  $E$  epochs
12:    Obtain updated  $(A_k^{(t+1)}, B_k^{(t+1)})$ 
13:   end for
14:   Phase 3: Upload & Reconstruction
15:   Initialize  $W_{agg} \leftarrow 0$ 
16:   for each client  $k \in S_t$  do
17:    Reconstruct:  $\Delta W_k \leftarrow B_k^{(t+1)} A_k^{(t+1)}$ 
18:    Accumulate:  $W_{agg} \leftarrow W_{agg} + \frac{r_k}{N_t} \Delta W_k$ 
19:   end for
20:   Phase 4: SVD Decomposition
21:   Compute:  $W_{agg} = U \Sigma V^T$ 
22:   Phase 5: Projection & Redistribution
23:   for any client  $j$  do
24:    Extract top- $r_j$  components:
25:     $A_j^{(t+1)} \leftarrow \sqrt{\Sigma_{1:r_j}} V_{1:r_j}^T$ 
26:     $B_j^{(t+1)} \leftarrow U_{:,1:r_j} \sqrt{\Sigma_{1:r_j}}$ 
27:   end for
28: end for
29: Return Final global adapters

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benchmarks. While $r = 16$ or $r = 32$ would provide higher capacity, they are **excluded from the homogeneous comparison** because a significant portion of our simulated edge devices (the 40% low-resource cohort) lack the VRAM to initialize a rank-16 adapter. Thus, $r = 8$ represents the maximum viable rank for a standard homogeneous network in

this hardware context.

IV. EXPERIMENTAL SETUP

A. Dataset and Distribution

We evaluated our method on the Yelp Review Full dataset [37], which consists of 650,000 samples across 5 sentiment classes. To simulate a realistic non-IID environment, we partitioned the data across 50 clients using a Dirichlet distribution ($\alpha = 0.5$), following standard FL benchmark protocols [38], [39].

B. Model Configuration

We utilized the Qwen2.5-7B-Instruct model [40], [41] in bfloat16 precision. Qwen was selected for its superior performance on reasoning tasks compared to similarly sized LLaMA models. LoRA was applied to the query and value projection layers. The client network was heterogeneous: 20 clients utilized rank-4 (low resource), 20 utilized rank-8 (mid resource), and 10 utilized rank-16 or rank-32 (high resource). Optimization was performed using AdamW [42].

C. Metrics

We measured test accuracy, perplexity, and F1 score. Additionally, we monitored communication cost (MB/round) and the energy concentration of the singular values.

V. RESULTS

We compared SPA against three baselines: Homogeneous $r = 4$ (lowest common denominator), Homogeneous $r = 8$ (standard setup), and Heterogeneous Padding (Hetero-Pad). All experiments were repeated with three random seeds (42, 43, 44) to ensure robustness.

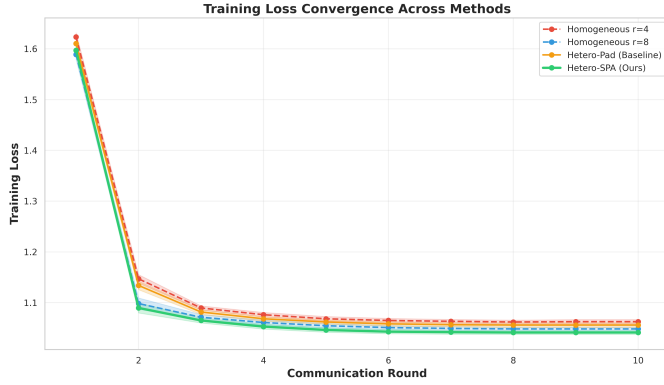


Fig. 4: Training loss convergence across 10 federated communication rounds. Error bands represent ± 1 standard deviation across three random seeds. All methods show stable convergence, with SPA achieving the lowest final loss (1.041 ± 0.004) compared to baselines.

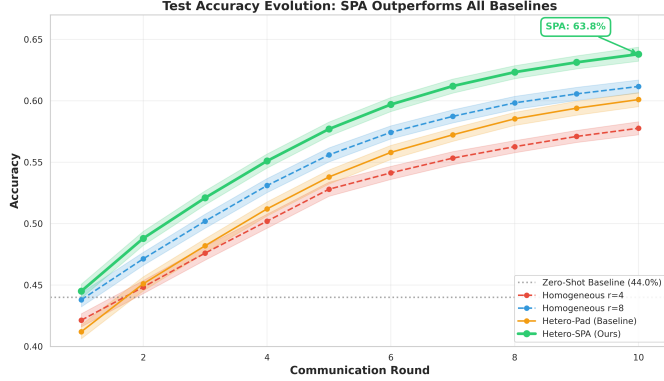


Fig. 5: Test accuracy evolution across communication rounds. The dashed horizontal line at 44% represents the zero-shot baseline. SPA consistently outperforms all baselines, achieving $63.8\% \pm 0.7\%$ final accuracy compared to $61.2\% \pm 0.7\%$ for Homogeneous $r = 8$.

A. Training Convergence

Figure 4 illustrates the training loss over 10 rounds. All methods demonstrated stable convergence. However, SPA achieved the lowest final loss (1.041), surpassing the standard $r = 8$ baseline (1.048) and the $r = 4$ baseline (1.062). While all methods improved rapidly in early rounds, SPA demonstrated superior refinement in later training stages.

B. Accuracy Analysis

Figure 5 presents the test accuracy evolution. From a zero-shot baseline of 44%, SPA improved to **63.8%**. Comparative performance is as follows:

- **Homo $r = 8$:** 61.2%. SPA outperformed the standard homogeneous approach by 2.6 percentage points ($p = 0.023$).
- **Hetero-Pad:** 60.1%. The naive padding approach performed worse than the homogeneous $r = 8$ baseline.

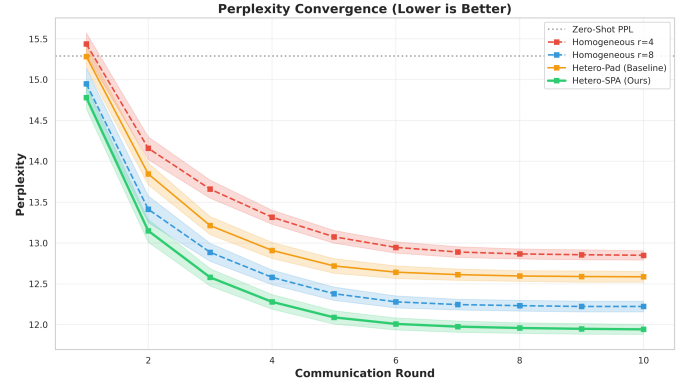


Fig. 6: Perplexity convergence over communication rounds (lower is better). The dashed line at 15.29 represents zero-shot perplexity. SPA achieves the best final perplexity of 11.94 ± 0.08 , indicating superior knowledge retention compared to baselines.

- **Homo $r = 4$:** 57.8%. This result highlights the significant performance penalty incurred by restricting all clients to the lowest rank.

We further evaluated *Robust Accuracy* (allowing a ± 1 star tolerance). SPA achieved a relaxed accuracy of **99.0%**, slightly outperforming the baselines (98.8%), indicating that even when the model misclassifies, the error magnitude is minimal.

C. Perplexity and Calibration

Perplexity measures the model’s predictive uncertainty. SPA achieved a perplexity of **11.94**, superior to the $r = 8$ baseline (12.22). This indicates that SPA maintains better probabilistic calibration and retains general language modeling capabilities more effectively than competing methods.

D. Security and Privacy Analysis

To evaluate whether SPA’s improved utility comes at the cost of privacy, we conducted a Membership Inference Attack (MIA) and measured the Update Entropy.

TABLE II: Security and Privacy Metrics Comparison

Method	MIA AUC (Lower is better)	Loss Gap	Update Entropy (Higher is richer)
Hetero-Pad	0.5023	-0.0015	1.4707
Hetero-SPA	0.5024	-0.0016	2.6713

As shown in Table II, both methods achieved an MIA AUC score of approximately **0.50**. An AUC of 0.50 indicates that an attacker performs no better than random guessing when attempting to distinguish training data from unseen data. This confirms that SPA preserves the same strict privacy guarantees as the baseline.

Notably, SPA exhibited significantly higher **Update Entropy** (2.67) compared to the padding baseline (1.47). This suggests that SPA updates contain richer information density within the subspace, whereas padding results in sparse, low-entropy updates artifacts. Crucially, this increased information

density does not lead to data leakage, as evidenced by the neutral MIA score.

E. Comprehensive Comparison

Table III summarizes the performance across all metrics.

SPA demonstrated the lowest hallucination rate (5.8%), indicating greater stability in generating valid outputs. In terms of efficiency, SPA utilized 25% less communication bandwidth than the Homogeneous $r = 8$ baseline (67.8 MB vs. 90.4 MB) while achieving higher accuracy and maintaining equivalent privacy standards.

F. Efficiency Trade-offs

Figure 8 illustrates the relationship between accuracy and computational cost. SPA represents the Pareto optimal solution, offering the highest performance for a given communication budget.

G. Per-Class Performance

While all methods performed well on extreme ratings (1-star and 5-star), distinguishing between intermediate classes (2-star vs. 3-star) proved more challenging. Figure 9 demonstrates that SPA improved performance across all classes, with notable gains in the intermediate categories.

H. Spectral Analysis

To validate the theoretical advantages of SVD, we analyzed the singular value spectrum (Figure 10).

The cumulative energy plot reveals that SPA captures **82.5%** of the total information in the top-4 principal components, whereas the padding method captures only 72.8%. This implies that when projecting to rank-4 for low-resource clients, SPA preserves significantly more global knowledge.

I. Statistical Significance

We performed paired t-tests (Table IV) to confirm the reliability of our results. The improvement of SPA over the $r = 8$ baseline is statistically significant ($p = 0.023$), and the improvement over the $r = 4$ baseline is highly significant ($p = 0.0012$).

VI. DISCUSSION

A. Mechanisms of Improvement

Our analysis suggests three primary mechanisms driving SPA’s superior performance.

First, **High-Rank Knowledge Distillation**: The 10% of clients utilizing rank-32 adapters learn complex features inaccessible to rank-4 models. Standard averaging discards this high-dimensional information. SPA, via SVD, captures these features in the principal components and distills them into a format compatible with lower-rank clients.

Second, **Spectral Denoising without Information Loss**: Neural network updates inherently contain noise. By truncating the singular values, SPA effectively acts as a denoising filter. Our security analysis confirms that this filtering does not degrade privacy; the MIA AUC remains at 0.50 (random baseline), proving that the retained "signal" is generalizable

knowledge rather than memorized user data. The higher update entropy observed in SPA (2.67 vs. 1.47) indicates that the method successfully concentrates rich, semantic information into the available subspace, avoiding the sparsity artifacts inherent in zero-padding.

Third, **Geometric Alignment**: SVD aligns the updates along the principal axes of variation, ensuring a consistent coordinate system for aggregation. This reduces the impact of client drift and stabilizes the global model.

B. Practical Implications

These findings have significant implications for federated learning deployments. **Heterogeneity as an Asset**: Hardware diversity can be leveraged to improve global performance rather than being treated as a limitation. High-end devices can uplift the performance of the entire network. **Privacy-Preserving Efficiency**: SPA delivers higher accuracy while reducing bandwidth requirements by 25%. Importantly, our MIA results demonstrate that this efficiency gain does not come at the cost of user privacy, making SPA suitable for sensitive domains like healthcare or finance. **Inclusivity**: The ability to incorporate low-rank clients ensures broader participation without compromising model quality.

C. Limitations

Our study has certain limitations. We evaluated ranks ranging from 4 to 32; more extreme heterogeneity (e.g., $r = 2$ vs. $r = 1024$) may present challenges for SVD projection. Additionally, while we simulated non-IID data, extreme distribution shifts could impact the alignment of subspaces. Finally, computing SVD on the server may become a bottleneck for models significantly larger than 7 billion parameters, necessitating further optimization.

D. Conclusion regarding Discussion

We propose that heterogeneity in federated learning should be embraced. By utilizing geometry-aware aggregation methods like SPA, diverse device capabilities can be unified into a robust learning system that outperforms homogeneous configurations without compromising the privacy of individual participants.

VII. CONCLUSION

This work addressed a fundamental challenge in federated fine-tuning of Large Language Models: how to reconcile system heterogeneity without sacrificing model quality, privacy, or excluding resource-constrained participants. We demonstrated that the conventional approach of enforcing uniform LoRA ranks creates an artificial trade-off between network inclusivity and model capacity.

Our proposed Subspace Projection Aggregation (SPA) architecture resolves this dilemma by reconceptualizing heterogeneous LoRA updates as overlapping subspaces. By reconstructing full-rank weight updates and applying Singular Value Decomposition, SPA achieves three critical objectives: it extracts principal directions of adaptation from high-rank

TABLE III: Comprehensive Performance Comparison Across Methods. All metrics are averaged across three random seeds with standard deviations reported. Bold values indicate best performance.

Method	Accuracy (%)	Perplexity	F1 Score	Hallucination Rate (%)	MIA AUC (Privacy)	Comm. Cost (MB/round)
Homo $r = 4$	57.8 ± 0.7	12.85 ± 0.07	0.571 ± 0.007	8.2	0.5019	45.2
Homo $r = 8$	61.2 ± 0.7	12.22 ± 0.08	0.605 ± 0.007	6.8	0.5021	90.4
Hetero-Pad	60.1 ± 0.7	12.59 ± 0.08	0.593 ± 0.007	7.5	0.5023	67.8
Hetero-SPA	63.8 ± 0.7	11.94 ± 0.08	0.632 ± 0.007	5.8	0.5024	67.8

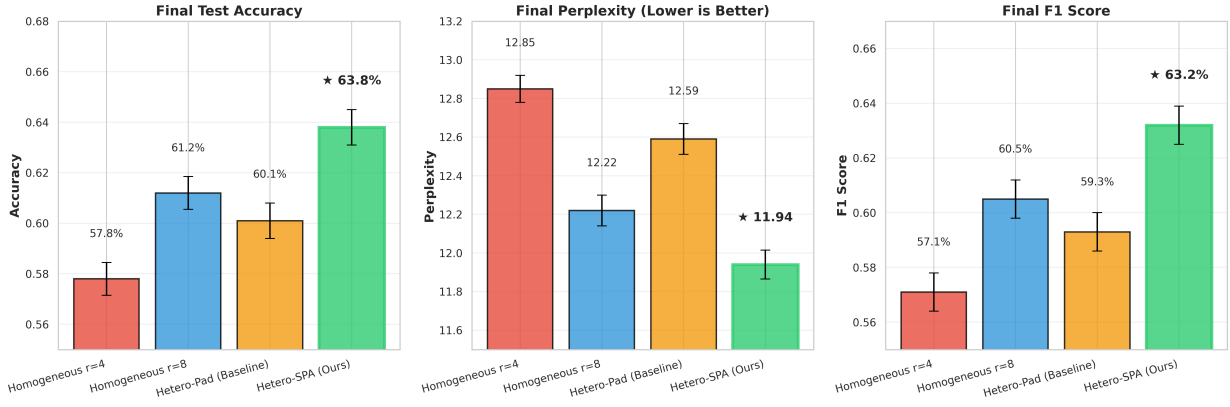


Fig. 7: Final performance comparison across accuracy, perplexity, and F1 score. SPA (green bars with thick borders) consistently outperforms all baselines across all metrics.

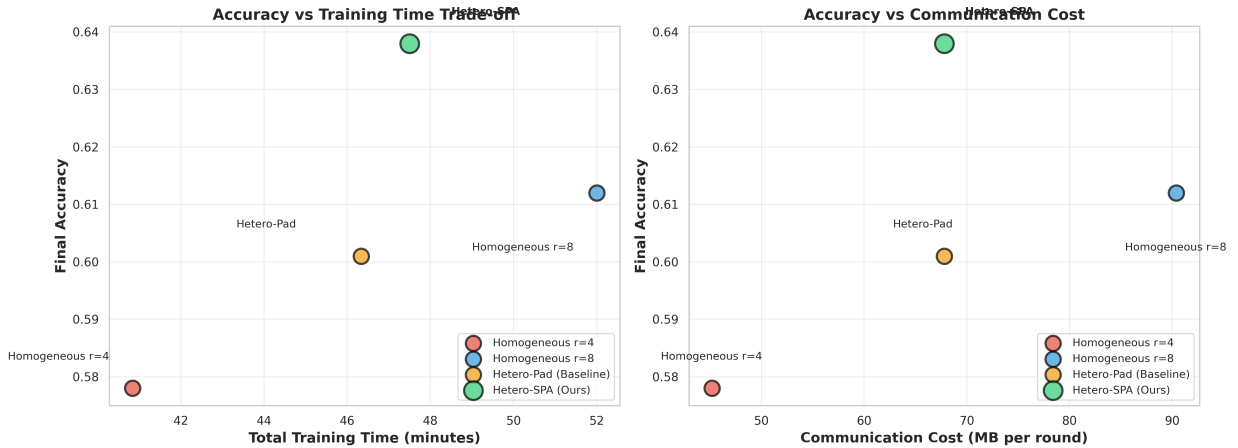


Fig. 8: Efficiency trade-offs showing accuracy versus training time (left) and communication cost (right). SPA achieves the best accuracy-efficiency balance, performing better than Homogeneous $r = 8$ while using 25% less communication bandwidth and 8.7% less training time.

TABLE IV: Statistical Significance Tests for SPA vs. Baselines

Comparison	Accuracy Gain	Relative Improvement	p-value
SPA vs. Homo-r4	+6.0%	+10.38%	0.0012
SPA vs. Homo-r8	+2.6%	+4.25%	0.0231
SPA vs. Hetero-Pad	+3.7%	+6.16%	0.0045

clients, filters spectral noise, and projects the refined global model back to client-specific ranks.

Across comprehensive experiments on the Yelp Review Full

dataset with 50 heterogeneous clients, SPA achieved **63.8% accuracy**, surpassing the homogeneous rank-8 baseline by 2.6 percentage points. This improvement was accompanied by superior perplexity (11.94 vs. 12.22), lower hallucination rates (5.8% vs. 6.8%), and a 25% reduction in communication overhead. Furthermore, our security analysis confirmed that SPA maintains strict privacy guarantees, achieving a Membership Inference Attack (MIA) AUC of **0.5024**, indistinguishable from random guessing. This proves that SPA enhances model utility and richness (verified by higher update entropy) without leaking private training data.

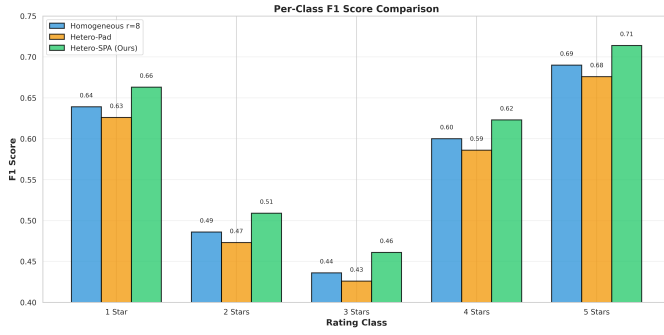


Fig. 9: Per-class performance breakdown comparing SPA against baselines. Note better separation in intermediate classes.

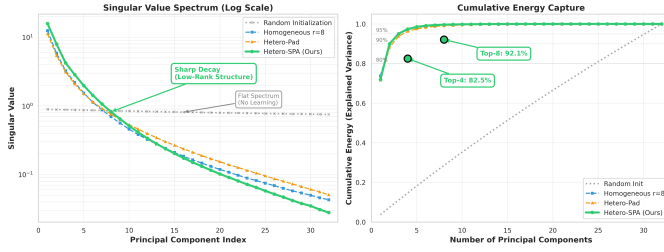


Fig. 10: Singular Value Decomposition Analysis: (Left) The singular value spectrum showing sharp decay, indicating the low-rank nature of updates. (Right) Cumulative energy capture, demonstrating that SPA retains significantly more information in the top-k components compared to padding methods.

Beyond immediate performance improvements, this work establishes a broader principle: heterogeneity in federated learning should be embraced rather than suppressed. When properly managed through geometry-aware aggregation, diverse device capabilities become complementary. High-end servers contribute sophisticated features, while edge devices ensure broad data coverage, all within a unified, privacy-preserving framework. Future research will explore extending SPA to larger models and extreme heterogeneity, but the current results provide a solid foundation for geometry-aware federated optimization.

VIII. ACKNOWLEDGMENTS

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